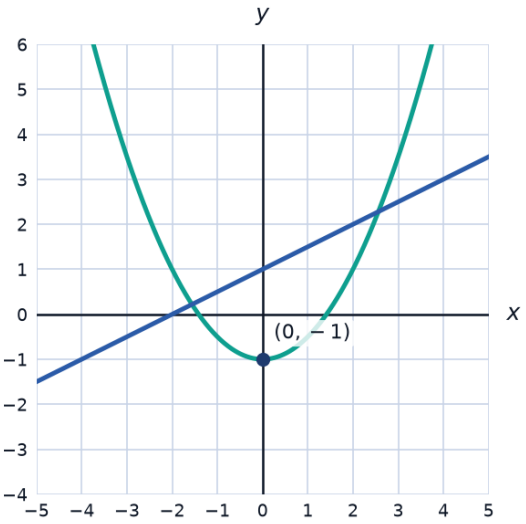
The Definition of the Derivative

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

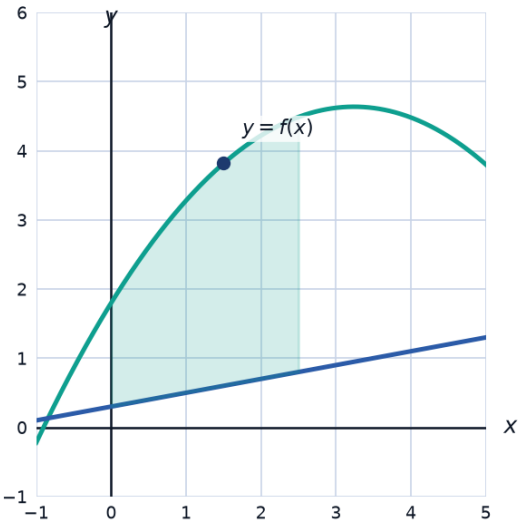
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use the Second Derivative Test to classify the critical points of $f(x) = 3x^3 - 4x$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

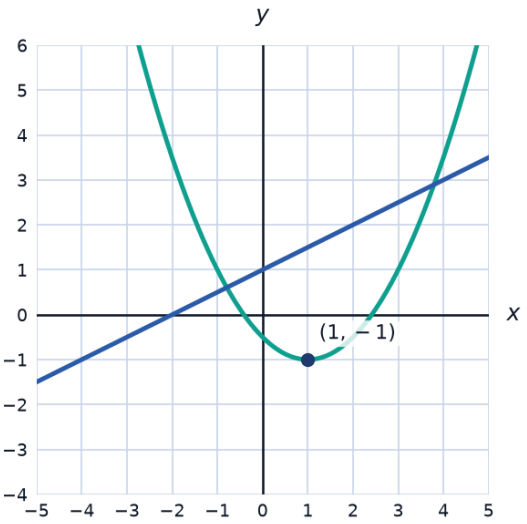
The graph of the function $f(x) = ax - 3$ is shown below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 0$ to $x = 4$ is 20 square units. Find the value of the parameter a.



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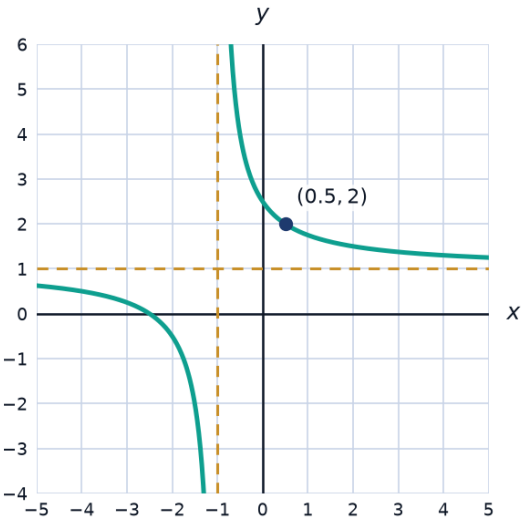
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds?



4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use L'Hopital's Rule to evaluate $\lim_{x \to \infty} \frac{7x}{e^{4x}}$.



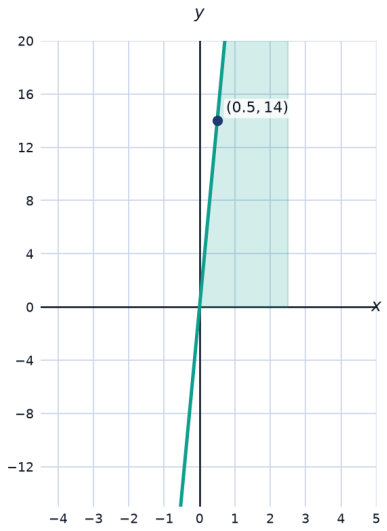
5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A 5-ft ladder leans against a wall. The bottom slides away from the wall at 2 ft/s. Find the rate at which the top of the ladder is sliding down the wall when the bottom is 3 ft from the wall.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the area of the region bounded by $y = 28x$ and $y = x^2$ in the first quadrant.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A differentiable function $f(x)$ is given by the equation $f(x) = x^2 - 6x + 10$. Find the instantaneous rate of change of $f(x)$ at $x = 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind the derivative of $f(x) = x^4 e^x - \frac{\sin x}{x^2 + 1}$.

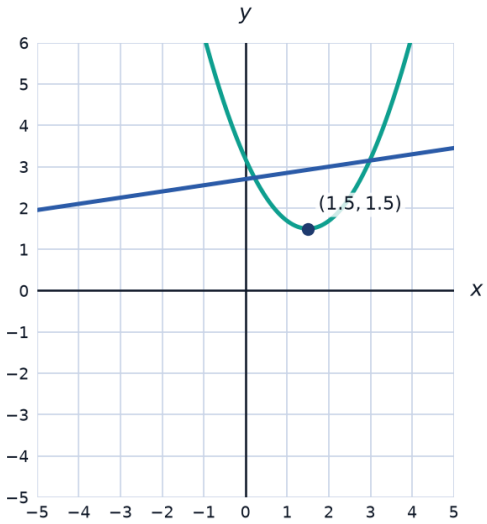
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedFind the 42nd derivative $f^{(42)}(x)$ of $f(x) = \sin(2x)$.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 2 · MediumLet f be a function whose derivative is $f'(x) = (x-1)^2(x+2)(x-4)$. Determine the x-coordinates of all relative extrema of f and classify each with justification.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 2 · MediumThe base of a solid is bounded by $y = \sqrt{x}$, the line $x = 9$, and the x-axis. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume of the solid.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumFind the value of a and b such that the line $y = 4x - 1$ is tangent to $f(x) = ax^2 + bx$ at $x = 1$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumFind the derivative $h'(x)$ for the accumulation function: $h(x) = \int_{x^2}^{\sin(3x)} \sqrt{1+t^4}\,dt$

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · AdvancedIf a student earns 32 out of 45 points on Section I and 34 out of 54 points on Section II, what is their composite score out of 108?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

[AP Calculus AB Exam Pacing & Scoring Strategy · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 The Chain Rule

[AP Calculus AB Exam Format & Units Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

[Unit 2 · Differentiation — Definition and Fundamental Properties · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Level 1 · FoundationsWhich single unit on the AP Calculus BC exam carries the highest individual weighting, and what specific topic within it is virtually guaranteed to appear on Section II as Free Response Question 6?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Level 3 · AdvancedEvaluate the limit: $\lim_{x \to 0^+} (\cos x)^{1/x^2}$

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Level 1 · FoundationsFind the area of the region bounded by $x = y^2$ and $x = y + 2$ using integration with respect to y.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which limit expression represents the derivative $f'(x)$ by definition?

a. $\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

b. $\lim_{h \to 0} \frac{f(x+h) + f(x)}{h}$

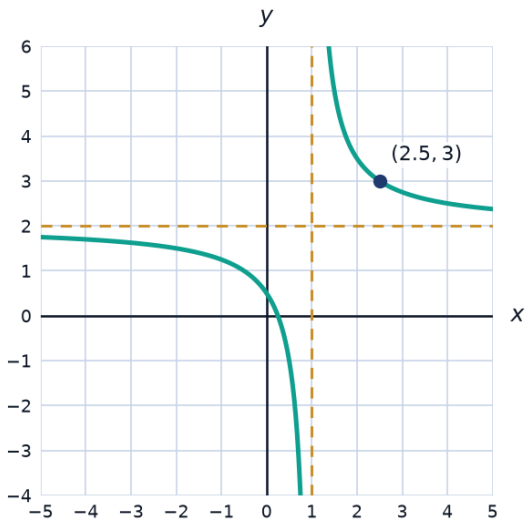
c. $\lim_{h \to \infty} \frac{f(x+h) - f(x)}{h}$

d. $\frac{f(x+h) - f(x)}{h}$

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which function and value of c are represented by $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$?



a. $f(x) = \sqrt{x}$ at $c = 4$

b. $f(x) = x^2$ at $c = 2$

c. $f(x) = \sqrt{x}$ at $c = 2$

d. $f(x) = \frac{1}{\sqrt{x}}$ at $c = 4$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a function f is differentiable at $x = c$, which of the following statements MUST be true?

a. f is continuous at $x = c$

b. $f'(c) > 0$

c. $f''(c)$ exists

d. f has a local extremum at $x = c$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why is the function $f(x) = |x - 3|$ not differentiable at $x = 3$?

a. The left-hand and right-hand derivative limits are unequal (corner)

b. $f(3)$ is undefined

c. f is not continuous at $x = 3$

d. $\lim_{x \to 3} f(x) = \infty$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $f(x) = x^{1/3}$ has what feature at $x = 0$ that prevents differentiability?

a. A vertical tangent line where $f'(x) \to \infty$

b. A jump discontinuity

c. A sharp cusp

d. A hole

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the equation of the tangent line to $f(x) = x^3 - 2x$ at the point $(2, 4)$?

a. $y - 4 = 10(x - 2)$

b. $y - 4 = 8(x - 2)$

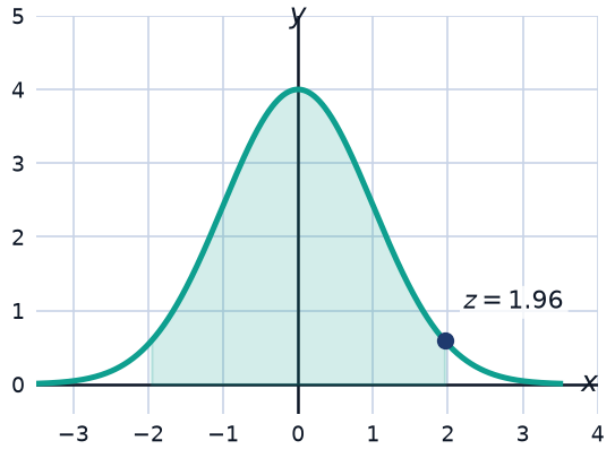
c. $y - 4 = 12(x - 2)$

d. $y = 10x - 12$

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the slope of the normal line to $y = x^2 + 3x$ at $x = 1$?



a. $-\frac{1}{5}$

b. 5

c. -5

d. $\frac{1}{5}$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

At what point on the parabola $y = x^2 - 4x + 3$ is the tangent line horizontal?

a. $(2, -1)$

b. $(2, 0)$

c. $(4, 3)$

d. $(0, 3)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given values $t = 2, 4, 6$ with $s(2) = 14, s(4) = 30, s(6) = 54$, estimate $s'(4)$ using the average rate of change over $[2, 6]$.

a. 10

b. 8

c. 16

d. 12

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the graph of f has a local maximum at $x = 5$ and is differentiable, what is $f'(5)$?

a. 0

b. 1

c. Positive

d. Undefined

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $f'(x) > 0$ for all $x \in (a, b)$, what does this indicate about the graph of f on (a, b)?

a. f is strictly increasing

b. f is strictly decreasing

c. f is concave up

d. f is linear

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A region R is bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. A solid is formed by revolving this region R about the y-axis. Find the volume of this solid. \\ \\ \\ (Hint: Consider the shell method and use a substitution that relates to the derivative of the denominator of the integrand.)

a. $\pi \ln(10)$

b. $2\pi \ln(5)$

c. $\pi \ln(5)$

d. $2\pi \ln(10)$

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The slope of the tangent to $y = x^3$ at $x = 2$ is:

a. 6

b. 8

c. 12

d. 16

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the derivative of $f(x) = 4e^x$.

a. e^x

b. $4e^x$

c. $-4e^x$

d. $4e^x + 4$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The table shows values of $g(x)$ at $x = [0, 3, 6, 8]$: $g = [2, 3, 8, 7]$. Using a right Riemann sum with these subintervals, approximate $\int_0^8 g(x)\,dx$.

a. 41

b. 47

c. 31

d. 50

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The table shows values of $g(x)$ at $x = [0, 2, 5, 7]$: $g = [7, 8, 12, 4]$. Using a left Riemann sum with these subintervals, approximate $\int_0^7 g(x)\,dx$.

a. 56

b. 65

c. 62

d. 60

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Using a left Riemann sum with 4 subintervals of equal width on $[0, 4]$, approximate $\int_0^4 (-x^2 + x + 4)\,dx$.

a. 5

b. -4

c. 2

d. 8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Using a midpoint Riemann sum with 4 subintervals on $[0, 8]$, approximate $\int_0^8 (-2x^2 + 1)\,dx$.

a. -333

b. -472

c. -326

d. -328

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The limit $\lim_{n\to\infty} \sum_{i=1}^{n} 3\left(\frac{4i}{n}\right)^2 \frac{4}{n}$ equals which definite integral's value?

a. 128

b. 64

c. 61

d. 32

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Using a right Riemann sum with 4 subintervals of equal width, approximate $\int_0^4 (x+2)\,dx$.

a. 16

b. 14

c. 20

d. 18

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Using a left Riemann sum with 2 subintervals of equal width, approximate $\int_0^6 (3x)\,dx$.

a. 81

b. 9

c. 23

d. 27

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1].

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right) dx$

c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right) dx$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2 + 1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$, and $g(x)$ is increasing.

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

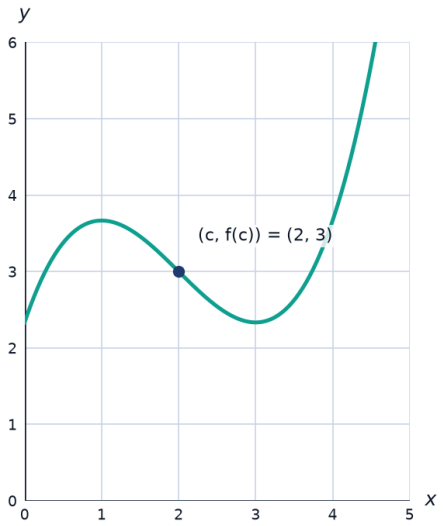
c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Area Between Curves

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.



a. 4

b. 3

c. 2

d. 1

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